

CSCI 721: Data cleaning and preparation

Week One: Overview

Outline for this week

- Course details
 - Textbook
 - Contact information
 - Course information and policies
- Lecture for Week One: Introduction to data cleaning and preparation
- Discussion questions

Textbook information

The textbook for this course is *Data Preparation for Data Mining* by Dorian Pyle, Morgan Kaufmann, 1999. The book is available on Books 24x7 through the RIT library. We will also be using some other texts and journal articles which I will post for you on MyCourses.

Contact information

The easiest (and fastest) way to contact me is through e-mail at cjr@cs.rit.edu.

If you are on campus, my office is GOL 3559.
On-campus office hours are Wednesdays from 10-11 a.m., and other times by appointment.

About the course

This course covers basic principles, algorithms, implementations, and applications of data cleaning and preparation. You should have access to basic office programs (word processing, spreadsheet, presentation software), and Adobe Acrobat Reader. You can download Reader [here](#) (it's free).

We will be using statistical methods in this class. Excel works for most statistical problems, but on occasion we may have to use other software, such as [R](#) or [Minitab](#).

Course policies and grading

Be sure to read the syllabus for complete information on topics and policies.

Grading evaluation is based on your participation in weekly discussions, completion of individual assignments, and performance on exams.

Course materials are available each Monday in MyCourses. You should familiarize yourself with the course software during this first week, if you have not used it before.

How the course works

- Read the weekly lectures and reference materials
- Do any graded homework assignments

Assignments and due dates are given in the lecture file.
Each assignment has its own MyCourses dropbox; usually, graded assignments are due Mondays at midnight.
- Participate in the discussions for that week. You must post your first response to the discussion question(s) by Wednesday. Then make sure you respond to two of your classmates' posts.
- Write in your individual blog. You will have a dedicated forum on MyCourses, that only you and I will see, to chronicle your progress (and ask questions or make comments) during the course.

How the course works, continued

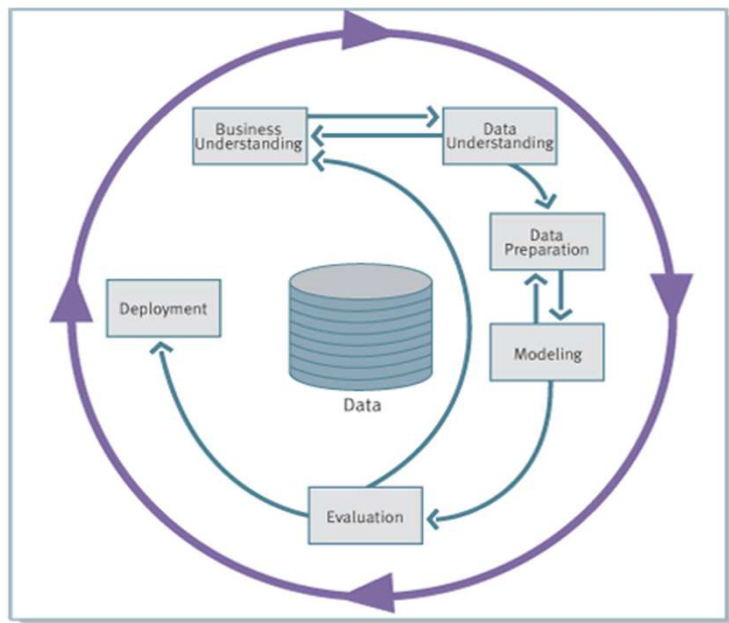
- Do the individual and group assignments. These aren't busy work. They are meant to help you practice the concepts we talk about in the lecture.
- To succeed in an online course, you need good time management skills since you don't have to physically go to class. What you get out of this class is directly related to what you put in.

Before class begins...

- Read the syllabus and post that you have read and understood the document.
- Check the activities list for weekly tasks and assignments.
- Read Chapters 1 and 3 in your text.

Intro to data preparation

The data mining process consists of several steps. If you've taken the data mining course already, this model is familiar to you.



The CRISP-DM* model shows the steps of the KDD (knowledge discovery in databases) process—beginning with defining the problem statement in the Business Understanding step.

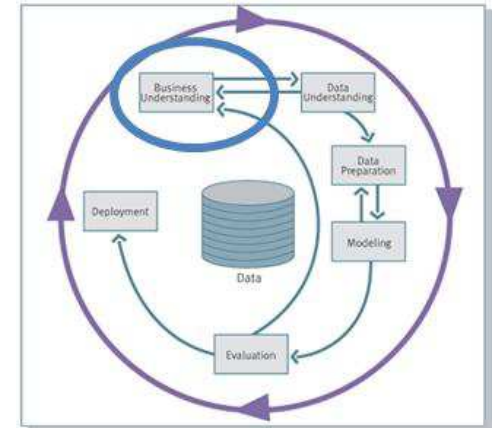
*Cross-Industry Standard Process for Data Mining

Business understanding

Business understanding encompasses

- Problem identification: what is the main issue driving the data mining effort?
- Resource identification: what people/ data/ hardware/ software are available for this project?
- Problem specification: what are the goals of this project? What would success look like?

These steps represent about 20% of the effort but contribute to 80% of the success.



Data understanding

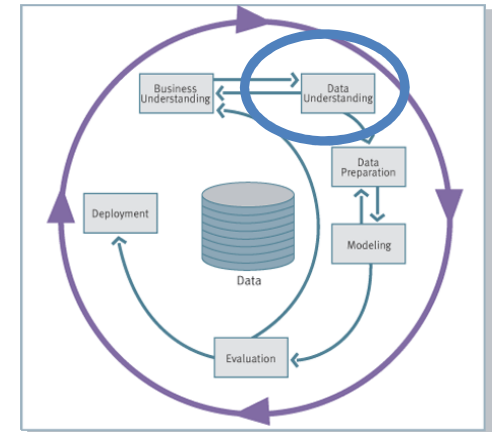
Data understanding encompasses

- Data prospecting

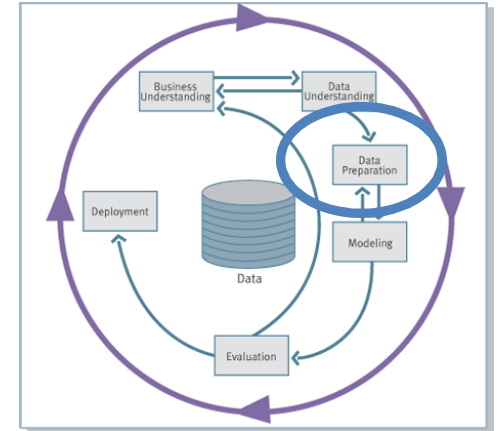
Data prospecting refers to the process of assembling the dataset (it may come from several different sources). Make sure you keep a log of where you got the data, what the attributes were, and what, if any, decisions you made about what to include or exclude.

- Domain knowledge elicitation

Domain knowledge elicitation means finding out from your domain expert what attributes are important, what the relationship is between attributes, and what would be interesting.



Data preparation



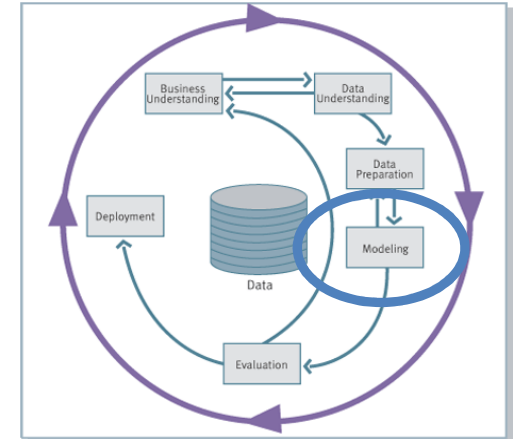
Data preparation refers to the process of cleaning, reducing, transforming, and formatting data for use by a data mining algorithm. This step, and the two preceding it, are the focus of this course.

Data preparation takes about 60% of the effort and is responsible for 15% of the success of a data mining project. So, between Business Understanding and Data Preparation, we've covered 80% of the effort and 95% of the success of a mining project.

Mining the data

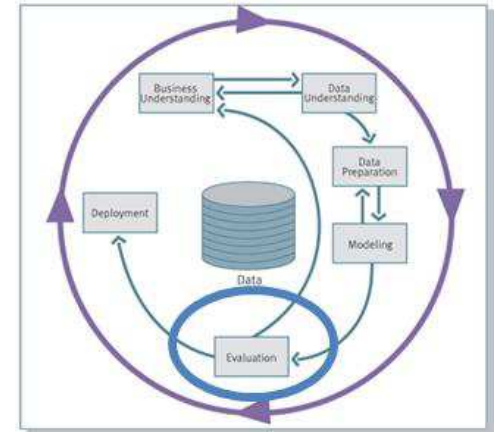
This step encompasses

- Data mining methodology identification
- Pattern discovery



The data mining method you choose is highly dependent on your goal and the user of the results. Different methods are used for different data mining tasks.

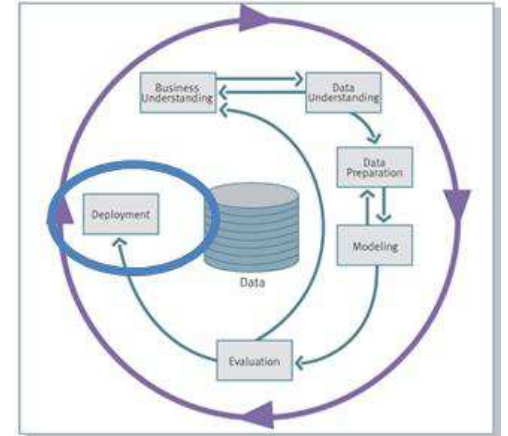
Evaluation



In this step, the mining model is validated and verified, and the knowledge produced is post-processed for the end user of the results.

Domain and data experts can help determine the usefulness of the model and the interestingness of the data mining outcome.

Deployment



This step includes reporting results to the end user, implementing the data mining model, and monitoring.

Although this model is somewhat linear, the actual process of developing a data mining solution is usually iterative and looping.

The data mining team

In order for a data mining project to be successful, we need input from three different experts: the **data** expert, the **domain** expert, and the **data mining** expert. Each has a specific and necessary role to play.

You'll have to play all three for this class; however, if you are involved in a data mining effort at a company or in research, be sure to make good use of the knowledge each one of these three experts offers.

The data expert can answer...

How was this dataset collected? Is it raw or aggregated data? Has the data been cleaned, summarized, or transformed in any way? What are the units of measurement? What is the range of this data? Where does it reside, and can we access it easily?

What other information can the data expert provide?

The domain expert can answer...

Are all variables equally important? What are the relationships between variables? What are the ranges of possible/likely values for each variable? Are there variables not captured that would be important? How should we deal with missing, empty, or incorrect values? What are the dependent variables?*

What other information can the domain expert provide?

*Target variables, or quantities being estimated or predicted.

The data mining expert can answer...

Which algorithm is the best fit for this task? Is there more than one algorithm that would be useful? What should the values of tuning parameters be to elicit the best result? What are the limitations of this algorithm relative to this task? How can we best prepare the dataset for mining and the results for analysis?

What other information can the data mining expert provide?

The data preparation process

In your text, the author makes the point that there's no magic software that perfectly prepares a dataset for mining. As we saw earlier, the data preparation step takes a long time and is critical to success of the data mining project. The step itself has its own process.

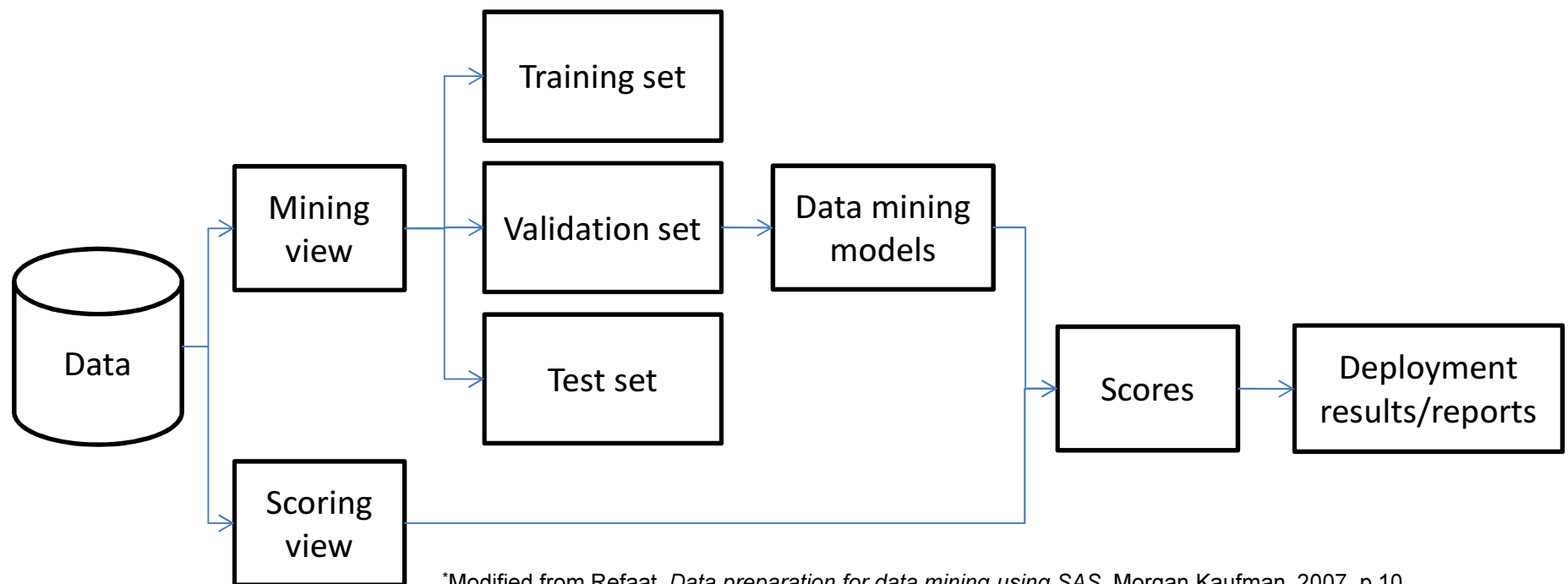
The author makes reference to a Planned Information Environment (PIE) and associated modules. We won't be using the software he wrote, but the concepts are still valid.

Steps in the data preparation process (according to Pyle)

- Accessing the data
- Auditing the data
- Enhancing and enriching the data
- Looking for sampling bias
- Determining data structure
- Building the PIE
- Surveying the data
- Modeling the data

A view of the data flow^{*}

You should remember that nothing about the knowledge discovery process is linear—so although this diagram shows a linear flow, in reality many steps may need to be repeated. The “mining view” here is analogous to Pyle’s PIE. The “scoring view” is similar to the mining view, but contains only the fields present in the data mining model, and does not contain the dependent variable. More about the scoring view later.



^{*}Modified from Refaat, *Data preparation for data mining using SAS*, Morgan Kaufman, 2007, p 10.

Accessing the data

Pyle discusses some issues with accessing data, but there are many more. When this book was written, mining was almost totally done on flat files (much data mining still is).

One of the issues he does not address, but which is becoming increasingly important, is that of distributed and heterogeneous data sources. What problems can you think of that would arise from combining data from disparate sources?

Auditing the data

Statistical data collection is rigorously designed to capture attributes of interest, with a particular range of values at a specific level of detail. In contrast, data mining datasets are often opportunistic samples that were not designed or collected with data mining in mind.

This reality means we are dealing with less-than-optimal data. Data preparation techniques can increase the utility of a dataset, but cannot turn lead into gold.

Enhancing and enriching the data

Sometimes we can aid the learning algorithms by supplying it with basic concepts that are present already in the dataset. In doing so, we avoid making the algorithm learn these concepts and let it focus on more sophisticated relationships.

For example, if we have the attributes TimeIn and TimeOut, we can create a new attribute called $\text{TotalTime} = \text{TimeOut} - \text{TimeIn}$. This new attribute explicitly models the relationship between TimeIn and TimeOut and spares the algorithm the trouble of learning it.

Sampling bias

Here again we see the effects of an opportunistic dataset. If we know the “state of the world” we can check to see if our dataset accurately reflects that state. Often, it does not. Sometimes, we can’t tell.

Sampling can introduce bias or attempt to correct for what’s already there. We’ll learn more about sampling methods later in the course.

Determining data structure

We would probably call the super- and macro-structure Pyle refers to *metadata*. Basically, that's data about the data. We need to know about the metadata so we can determine the best ways to handle the different attributes and the quality problems they may present.

Building the PIE

Pyle's PIE is, essentially, an automated version of the log file you should always keep when you make changes to a dataset. The PIE ensures that these changes can be applied to any dataset from the same source with the same attributes and the same data mining goal. Data mining is rarely a one-off project, and it's certainly not linear. The automated PIE allows you to model and test other subsets of the data.

Surveying and modeling the data

The last two steps begin to turn the transformed dataset into actionable models, the whole point of the data mining exercise.

Why do we have to do data prep?

If we just go ahead and analyze the data in its raw form, we open ourselves up to several problems, and our results are not going to be useful. Clean, prepared data helps avoid GIGO (garbage in, garbage out).

What are some of the problems we might encounter? Dasu and Johnson* list four of the most likely:

1. Spurious results
2. Misplaced faith in black boxes
3. Limitations of popular models
4. Buyer beware—no guarantees

*Dasu, T. and Johnson, T. *Exploratory Data Mining and Data Cleaning*, New Jersey: Wiley and Sons, 2003, pp 2-3.

Spurious results

Datasets often include errors generated by the data collection process—such as signal drop-outs and delays, or spelling errors. For example, Dasu and Johnson found errors in a telecommunications dataset that, if not removed, would have generated patterns of signature usage that were not accurate. The model built from those patterns would then have been incorrect.

Misplaced faith in black boxes

Data mining is more than clicking buttons on a piece of software. It's not enough to know how an algorithm works, you also need to know how good the data is that you are presenting to the algorithm. You also need to know about the process that generated that data, and what constitutes good data.

Otherwise, you can't validate your results.
Garbage in, garbage out.

Limitations of popular models

Interestingly, a survey of data miners showed that people tend to choose a model based on the available software—not whether the model was best for the dataset and the data mining goal.

They also tend to choose the type of model they understand best, even if basic assumptions of the model (such as the normally distributed assumption of linear regression) do not hold.

Another limitation is scalability; some methods, like clustering and neural networks, may be too slow for a particular dataset.

Buyer beware—no guarantees

How do we know if our model fits the underlying process? Many data mining models, such as clustering, do not have goodness-of-fit measures and validation is difficult.

We may also find correlations between variables that are spurious (like the correlation between ice cream consumption and drownings in the summertime, or between the S&P index and butter production in Bangladesh). Correlation does not necessarily mean causation.

Avoiding the pitfalls

Cleaning and understanding the data will help us in our efforts to avoid these problems while delivering a robust result. Remember that data cleaning, like the KDD process itself, is not linear. You may have to go back to earlier steps and revise the dataset, as you learn more about its contents.

Summary

In this course, we'll explore ways to maximize the return on investment in the data mining process by ensuring we are presenting the best possible data to the learning algorithm. There's no substitute for learning as much as we can about our data, and making informed decisions to mitigate its problems and challenges.

While processes may seem boring and it's tempting to skip steps, a careful approach to data preparation is critical to success.

Discussion for this week

Discussion questions:

- What problems can you think of that would arise from combining data from distributed and heterogeneous sources? How would you go about mitigating or eliminating those problems?
- Pyle's book was written in 1999. What issues do we face now with data mining that weren't apparent 15 years ago?

Assignments for this week

- Discussion assignments: due by 2/2
 - Read the syllabus and post your confirmation in the appropriate forum.
 - Post your introduction in the introduction forum.
 - Answer the discussion question and respond to your classmates' posts.
- Don't forget to form your project group and start thinking about the research paper.